Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/13/2006
PI Name:	Clark, Noel A. Ph.D.		
Project Title:	Structure and Dynamics of Freely Suspended Liquid Crystals		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	noel.clark@colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-941-1008
Organization Name:	University of Colorado		
PI Address 1:	Physics Department, CB390		
PI Address 2:	2000 Colorado Ave		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0001	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Griffin, DeVon	**Contact Phone:**	216-433-8109
Contact Email:	DeVon.Griffin@grc.nasa.gov , DeVon.W.Griffin@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: funds reinstated until 11/30/2007, per J. Cohen (HQ) 8/2006. [Note - end date changed from 11/30/2003 grant renewed for 4 yrs/new NAG#]		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Maclennan, Joseph (University of Colorado) Glaser, Matthew (University of Colorado)		
Grant/Contract No.:	NNC04GA50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

We pursue ground-based studies of freely suspended liquid crystal films with an eye toward µg experiments, as well as atomistic computer simulation of liquid crystal phases, probing the molecular origins of observed film behavior. Ground-based studies focus on tethered smectic bubbles on which islands (isolated pancake-shaped thicker regions) can be generated and observed. The islands form a novel interacting particle array confined to the two-dimensional surface of the bubble. We observe islands (using reflected light microscopy) and manipulate them (using optical tweezers) and carry out computer simulation of the interaction of islands on contact, in an effort to understand their stability against aggregation.

Our work is directed toward developing a deeper understanding of condensed phases under conditions of reduced dimensionality. Films are locally optically anisotropic so that the pattern of molecular orientation within the film plane can be effectively studied by depolarized reflected light microscopy are also being studied. In particular, topological defect motion can be observed and the simplest liquid crystal defect dynamical process, the attraction and annihilation of oppositely charged, unit-strength defects (vortex-antivortex pairs), probed. Experiments to date reveal significant shortcomings in our understanding of this process. Application of in-plane electric fields to tilted smectic films reveals a host of interesting field-induced structures and transitions, and leads to novel methods for probing film structure. For example, by comparing the response of even layer numbered and odd layer numbered films, basic information about the nature of the polar ordering (ferroelectric or antiferroelectric) of the smectic can be obtained. Our recent discovery of the spontaneous formation of chiral domains in fluid smectic phases of achiral banana-shaped molecules (the first observation of spontaneous chiral domain formation in a fluid since Pasteur's discovery in crystals 150 years ago), based on observation of free film response to applied electric field, will be pursued, as will the recent discovery of freely suspended filaments of smectics comprised of banana-shaped molecules.

Smectic liquid crystals are phases of rod shaped molecules organized into one dimensionally (1D) periodic arrays of layers, each layer being a two dimensional (2D) liquid between one and two molecular lengths thick. This inherent fluid-layer structure and the low vapor pressure of smectic liquid crystals enable the long term stabilization of freely suspended, single component, fluid films quantized in integral numbers of layers in thickness, down to a single molecular layer (30Å). These ultrathin freely suspended liquid crystal films are structures of fundamental interest in condensed matter and fluid physics. They are the thinnest known stable condensed phase fluid structures and have the largest surface-to-volume ratio of any stable fluid preparation, making them ideal for the study of the effects of reduced dimensionality on phase behavior and on fluctuation and interface phenomena. Their low vapor pressure and quantized thickness enable the effective use of microgravity to extend the study of basic capillary phenomena to ultrathin fluid films.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Industry Affiliates: Instec, Inc., Boulder, CO, Imaging Technology International, Boulder, CO
Innovative Technologies Developed: development of ultrathin bubbles, dispersion system of nanodroplet of liquid crystals, manipulation technique of nanodroplet of liquid crystals

Research Audience: physicists, chemist, engineers, materials scientist

Graduate Student Employment: Nattaporn Chattham - Assistant Professor at Kanseisart University in Bangkok, Thailand

Research Impact on American/Earth Benefits: fundamental studies of fluid dynamics and self-organization in two dimensions

Task Progress:

INTRODUCTION: Smectic liquid crystals composed of fluid layers can be made to form freely suspended, single component, fluid films as thin as a single molecular layer (3nm). These films are of fundamental interest in condensed matter physics. They are the thinnest known stable fluid structures and have the largest surface-to-volume ratio of any fluid preparation, making them ideal for studies of two dimensional phase transitions, fluid flow, and fluctuation and interface phenomena. They offer unique opportunities to effectively use microgravity to extend the study of basic capillary phenomena to ultrathin fluid films. The smectic layering forces films to be an integral number of layers thick, quantizing their thickness in layer units and forcing a film of a particular number of layers N to be physically homogeneous with respect to its layer structure over its entire area. This inherent fluid-layer structure and the low vapor pressure of smectics enable, in a straightforward way, the stabilization of freely suspended, single component, layered fluid films for detailed study. The main objective of the research supported by this Grant is to advance the understanding of fluid physics by exploiting the unique characteristics of freely suspended liquid crystal films in a microgravity environment. These experiments focus on use of a novel collective system of interacting, one dimensional interfaces in two dimensional fluids for the study of the fluid physics in two dimensions and the capillarity of ultrathin fluid films. We propose to pursue the definition of experiments on island generation, observation (using reflected light microscopy), and manipulation (using optical tweezers) for the µg environment.
PROGRESS: "INTERACTION OF ISLANDS ON SMECTIC C FREELY SUSPENDED FILMS." We have directly measured the elastic dipolar interactions between islands on freely suspended smectic films using the multitrap optical tweezer technique. The interactions between islands in non-chiral materials are surprisingly small but they increase significantly in chiral materials. This is manifest under the microscope as an increase in the amount of island chaining.

The measured interaction potentials can be fitted qualitatively with two dimensional energy curves computed assuming that the interaction is purely elastic in nature. Measurements of the attractive force at fixed island separation as a function of enantiomeric excess, providing direct confirmation that the interac-tions increase with increasing chirality. In chiral films, the elastic constant so obtained is significantly (~10 times) greater than the values typically found in non-chiral materials. In summary, we have observed that the magnitude of the interaction force between islands is much smaller in the racemic smectic C case than in the chiral smectic C*, an effect we attribute to long-range coulombic forces arising from polarization charges.

PROPOSED PLANS: 1) Studies of Island Interactions in High Polarization Materials; As we have shown above, the interaction force between islands on a freely suspended film depends quadratically on spontaneous polarization charge. This means that both elastic interactions and polarization charge effects have to be considered in order to model the interactions between islands.

We propose to continue to use multiple optical trapping techniques to study island interactions and to observe the c-director (projection of molecular long axis onto the layer plane) and polarization orientation in high polarization SmC films. R. Meyer recently described a textural transformation in islands from pure bend to reversing spiral due to a strong correlation between the spontaneous electrical polarization of the ferroelectric phase and the apparent difference between the bend and splay elastic constants. In this study, we will continue to study textural transformations both inside (+1 defect) and outside (-1 defect) islands on very high polarization (>200 nC/cm2) films. Another goal is to develop simulation techniques to be able to compute the c-director orientation on high polarization SmC* films. We plan to calculate the effective interaction between two disk-defect pairs using a two-dimensional elastic free energy with different contributions for splay and bend deformations and to include the Coulomb energy due to polarization charges on the film.

2) Island Generation via Inkjet Printing; As we have described in earlier reports, islands generated spontaneously on films or bubbles, or those created using shear flow, show a wide range of diameters and thicknesses in any given experiment. We have nevertheless been able to study island interactions between carefully selected pairs (or clusters) of such “natural” islands that are of similar size and layer number.

With a view to generating large numbers of islands of uniform size, we have recently performed preliminary experiments using an inkjet printer head to deposit homogeneous droplets of low viscosity fluids on thin smectic films. When drops of water, for example, adhere to the film, they nucleate new smectic islands. Liquid crystal material from the film wets the water and covers the free surface of the droplet over the course of several minutes, until the remains of the original drop (which will have shrunk by evaporation during this time) are completely covered by smectic material. Such islands are very uniform in size and interact, forming chains similar to those observed previously. They may exhibit c-director textures that are more complex than those seen in “natural” islands. We propose to continue to develop inkjet techniques as a method for delivering droplets of liquid crystal. We will investigate whether we can print small volumes of nematic or smectic material onto films in order to generate islands directly. To compensate for the higher viscosity of smectics, the reservoir and print head may need to be heated. It is known from work by Meyer and Pieranksi that dust or smoke particles can be used to nucleate smectic islands. One of our goals is to use a relatively volatile liquid (such as water or alcohol), or liquid crystal itself, to create monodisperse islands that have no remnant impurities in them and hence will be fully equivalent in all their physical properties to “natural” smectic islands.

Bibliography Type:	Description: (Last Updated: 12/04/2024)
Abstracts for Journals and Proceedings	Clark NA. "Liquid crystal phases of molecular bananas: polarity and chirality as broken symmetries." Presented at the American Physical Society Meeting, Baltimore, MD, March, 2006. Bulletin of the American Physical Society 2006 Mar;51: B1. , Mar-2006
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Polarization-enhanced interaction between islands on freely-suspended smectic C* films." Presented at the American Physical Society Meeting, Baltimore, MD, March, 2006. Bulletin of the American Physical Society 2006 Mar;51:U21. , Mar-2006
Abstracts for Journals and Proceedings	Park CS, Clark NA. "Dynamics of dislocation loops in one and two layer freely suspended films of fluorinated smectic a liquid crystals." Presented at the 21st International Liquid Crystal Conference, Keystone, Colo., July 2-7, 2006. 21st International Liquid Crystal Conference Abstracts, abstract SURFP-45, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Polarization-enhanced interaction between islands on freely-suspended smectic C* films." Presented at the 21st International Liquid Crystal Conference, Keystone, Colo., July 2-7, 2006. 21st International Liquid Crystal Conference Abstracts, Abstract DFCTP-12, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Optical tweezer studies of liquid crystals using optical multiple traps." Presented at the American Physical Society Meeting, Los Angeles, CA, March 2005. Bulletin of the American Physical Society 2005 Mar;50: 221P. , Mar-2005
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Direct measurement of dipolar elastic interaction forces between islands on freely suspended smectic C films using multiple optical tweezers." Presented at the 10th Conference on Ferroelectric Liquid Crystals, Stare Jablonki, Poland, Sep. 12- 17, 2005. 10th Conference on Ferroelectric Liquid Crystals Abstracts, September 2005. , Sep-2005
Articles in Peer-reviewed Journals	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Manipulation of disk-shaped islands on freely suspended smectic films and bubbles using optical tweezers." Ferroelectrics 2004 Dec;310:131-5. http://dx.doi.org/10.1080/00150190490510537 , Dec-2004
Articles in Peer-reviewed Journals	Park CS, Clark NA, Noble RD. "A convective turbulent state that spatially orders upon increased drive." Physics of Fluids 2005 May;17(5)055101. http://dx.doi.org/10.1063/1.1886205 , May-2005
Articles in Peer-reviewed Journals	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Direct measurement of interaction forces between islands on freely suspended smectic C films using multiple optical tweezers." Ferroelectrics. 2006 Dec;344:71-80. http://dx.doi.org/10.1080/00150190600966862 , Dec-2006
Articles in Peer-reviewed Journals	Link DR, Chattham N, Maclennan JE, Clark NA. "Effect of high polarization on defect structures and orientational dynamics of tilted chiral smectic freely suspended films." Physical Review E Statistical, Nonlinear, and Soft Matter Physics. 2005 Feb;71(2):021704. http://dx.doi.org/10.1103/PhysRevE.71.021704 , Feb-2005
Awards	Clark NA. "2006 Oliver E. Buckley Condensed Matter Prize, March 2006." Mar-2006